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An overview of microplastic in marine waters: Sources, abundance, characteristics and negative effects on various marine organisms

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ABSTRACT

In this study, the literature has been reviewed for the following purposes: (1) to summarize land-based and marine-based sources of microplastics (MPs) in the marine environment, (2) to evaluate the abundance of MPs in marine waters and factors affecting MP abundance in marine waters, (3) to review the general polymer types, shapes, sizes and colors of MPs in marine waters and to determine the dominant characteristics of MPs in marine waters, (4) to reveal the negative effects of MPs on various marine organisms (phytoplankton, zooplankton, corals, fish, sea turtles, seabirds). This present review provides an overview of the sources, abundance, and characteristics of MPs in the marine environment, which may guide future marine plastic/MP pollution management strategies and the improvement of MP removal methods from marine waters. Additionally, this review highlights the presence of MPs in various organism in the marine environment and the negative effects of MPs on marine organisms.

1. Introduction

Features of plastics such as ease of manufacture, lightness, flexibility, durability, resistance to moisture and chemicals, and low-cost increase the production and consumption of plastics. Global plastic production reached 390.7 million tons in 2021, 90.2% of which were fossil-based plastics [1]. It is estimated that in 2050, the accumulation of macroplastic and microplastic (MP) in the environment will be 2.2 and 3.1 gigatonnes, respectively [2]. Today, packaging and building-construction applications are the two plastic markets with the largest percentage in the distribution of global plastic use by application [1]. Plastic pollution has become an environmental threat on a global scale due to the increasing population, the increasing demand for plastic consumption, the lack of laws and regulations for waste management, the lack of a high rate of recycling of plastic waste in many parts of the world, the lack of implementation of effective waste management strategies, and lack of environmental consciousness [3]. Plastics can remain in the environment without degrading for hundreds of years. In addition, harmful chemicals added to plastics to increase their physical and chemical performance are also released into the environment along with plastics [4]. Moreover, MPs and nanoplastics, which are formed by breaking down plastics in the environment into smaller pieces, have become a threat to terrestrial and aquatic ecosystems.

MPs are plastic particles smaller than 5 mm in size. Considering their sources, they are divided into two groups: primary MPs and secondary MPs. Primary MPs are small-sized and relatively smooth-shaped MPs produced for commercial use. Microbeads used in personal care products and cosmetics (toothpaste, exfoliating face washes, eyeliners, soaps, etc.), and pellets used in the plastic production process are included in the primary MPs group [3]. Secondary MPs are formed as a result of the breakdown of macroplastics and mesoplastics into particles smaller than 5 mm in size by biological, chemical, and physical factors [3]. For instance, secondary MPs are formed by breaking plastic shopping bags, food packaging, water bottles, bottle caps, automobile tires, synthetic clothes, pipes, and shoe soles into smaller pieces. MPs are present in surface water [5], groundwater [6], wastewater treatment plants [3], drinking water treatment plants [7], soil [8], air [9], landfill leachate [10], bottled water [7,11], tap water [7,11], in other words, everywhere.

Plastics in the marine environment cause many negativities by attaching to organisms, reducing their mobility, causing death, and negatively affecting organisms living in benthos by accumulating in sediments [12]. In the marine environment, MPs are ingested by marine organisms and simultaneously cause many negative effects on their vital functions (various negativities in their growth, reproduction, diet, gastrointestinal tract, immune systems, etc.) [13–15]. In addition, MPs and the additives they contain are transferred through the food chain from the lower trophic level to the upper trophic level in the marine environment and finally transferred to humans [16]. Therefore, every process, from the sources of MPs entering the marine ecosystem to their effects on marine organisms, should continue to be investigated by researchers and monitored meticulously.

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In the first part of this review, land-based and marine-based sources of MPs in the marine environment were included. Then, the abundance and characteristics (polymer type, shape, size and color) of MPs in marine waters were discussed, respectively. In the last part of the review, the results of studies investigating the occurrence frequency, abundance, and characteristics (polymer type, shape, size and color) of MPs in various marine organisms were summarized. Additionally, studies examining the negative effects of MPs on various marine organisms from the lower trophic level to the upper trophic level in the marine environment were discussed and the negative effects of MPs were summarized.

2. Sources of microplastic in the marine environment

Increasing plastic consumption, poor waste disposal and recycling strategies, and lack of lack of environmental consciousness are making plastic pollution worse in the world. Since air, water, and soil environments always interact with each other, plastic particles in the environment can be transported from one environment to another in different ways. MPs are transported into the marine environment from land-based and marine-based sources, causing serious damage to the marine ecosystem. In an article published in 2014, it was reported that at least 5.25 trillion plastic particles were floating in the world's oceans (Eriksen et al., 2014).

Since almost half of the world's population lives within 50 miles of the coast, terrestrial plastic waste reaches the coastal region and approximately 80% of the plastic in the marine consists of land-based sources [17,18]. Land-based plastics/MPs result from poor waste management and disposal [19], landfilling [19], effluent of wastewater treatment plants [20], and runoff from streets and agricultural areas [21]. Land-based plastics include plastic bags, plastic bottles and caps, food and beverage packaging, cigarette butts, and many other plastic items used in daily life. Land-based plastics are plastics that people leave on the beach or plastics that are not disposed of properly and eventually end up in the marine environment [22]. These types of plastics are transported into the marine environment by rivers, precipitation, wind, tides, and floods [23]. For instance, it has been reported that 1.15–2.41 million tons of plastic waste enters the oceans from rivers every year and most of this input is from Asia [24]. Extreme natural events, such as high winds, hurricanes, storms, heavy rains, and tsunamis, cause large amounts of land-based plastic to enter the marine environment [25]. Additionally, land-based air-borne MPs can travel tens of kilometers and dominate the marine boundary layer and MPs can enter the atmosphere from the marine or, conversely, from the atmosphere to the marine [9].

Open dumpsites are areas where waste is stored randomly. Open dumpsites cause visual pollution, odor problems, easy spread of pathogens, and explosions, as well as surface water and groundwater pollution. Particularly in open dumps, when precipitation comes into contact with the waste, suspended and dissolved substances in the waste pass into the leachates, and leaches that enter surface waters such as seas and oceans through surface runoff pose a threat to the aquatic ecosystem [26]. On the other hand, sanitary landfilling is one of the final disposal methods that involves the controlled storage and management of solid waste in areas built according to engineering principles and containing natural and/or synthetic materials with very low permeability [27]. Plastic wastes in sanitary landfills are broken down into smaller plastic particles such as MPs by physical, chemical, and biological effects [28]. Studies have revealed that MPs in different polymer types, shapes, and sizes are found in leachates [10,19,29]. In the sanitary landfilling process, it is essential to collect the leachate generated in landfills in a controlled manner and treat it in a way that does not harm the environment. Discharge of leachate, which has a high polluting effect and is very difficult to treat, without being effectively treated, also invites the entry of MPs into the marine environment.

MPs that pass into wastewater as a result of washing synthetic clothes [30] and fabrics and using personal care products [31] come to wastewater treatment plants. However, the preliminary, primary, secondary, and tertiary treatment stages in domestic and/or industrial wastewater treatment plants, where physical, chemical, and biological processes are applied, cannot ensure the removal of MPs with 100% efficiency [3]. Studies have shown that MPs are present even in the permeate of membranes with very small pore sizes, such as reverse osmosis (RO) membranes used in wastewater treatment plants [32,33]. Therefore, it shows that even membrane technologies capable of removing MPs larger than the membrane pore size with high efficiency, regardless of polymer type, density, and shape, cannot completely remove MPs from wastewater [34]. Deterioration of membrane integrity, low membrane hardness and damage to the membrane material, especially by MPs with sharp corners, and the release of particles from the polymeric membrane structure are among the possible reasons for the presence of MPs in the permeate of membranes [34]. Many researchers have reported that millions of MPs are released by discharging effluents of WWTPs into surface waters [35–37]. Considering that even wastewater containing large amounts of treatment processes causes a significant amount of MP to enter aquatic environments, it can be estimated that the impact of untreated and uncontrolled domestic and industrial wastewater on MP pollution in aquatic environments is much greater [38–40].

Surface runoff from rural and urban areas and agricultural areas is an important way of transporting MPs from the terrestrial environment to rivers, lakes, estuaries, seas, and oceans. It has been reported that the global concentration of MPs in urban run off varies between 0–8580 particles/L [41]. MPs in worn tire particles, worn road marking paints, artificial grass, cigarette butts, and city dust are carried to surface waters by surface runoff [3]. The contribution of tire wear and tear to global plastic pollution in the oceans is estimated to be around 5%− 10% [42].

Many studies have proven the presence of MPs in soil [8,43,44]. Mulching to increase crop productivity in agricultural areas contributes significantly to MP pollution in the soil. For instance, a recent study by Zhang et al. (2021) showed that the abundance of MP in mulched agricultural soils ranged from 240–2253 items/kg (average: 754 ± 477 items/kg), while the abundance of MP in un-mulched agricultural land soils ranged from 173–800 items/kg (average 376 ± 149 items/kg) and MP pollution in the soil increased with long-term mulching [45]. In addition, since the sludge of wastewater treatment plants contains high amounts of MP [3,37] the use of WWTP sludge as agricultural fertilizer also causes an increase in the abundance of MP in the soil. Therefore, MPs, especially in agricultural areas where mulching is applied and WWTP sludges are used as agricultural fertilizers, can significantly contribute to the MP input in the marine environment through surface runoff.

A significant portion of plastic pollution in the marine environment consists of land-based plastics (~80%), while the remaining ~20% consists of marine-based plastics [46]. Marine-based plastics originate from fishing-related activities (fishing gear, nets, ropes, etc.), aquaculture, ship activities, and offshore industries [46]. Fishing gears used in fishing and aquaculture are generally made of polymers such as polyethylene (PE), polypropylene (PP) and polyamide (PA) [47]. Therefore, the uncontrolled release of plastic ropes and nets into the marine environment and their falling, rupture, wear, and tear cause marine MP pollution. Syversen and Lilleng (2022) estimated that 208 tons of MP are produced annually from the Norwegian fishery, while the number of MP from fishing worldwide is 4622 tons per year. Plastic waste and wastewater released from ships into the marine environment, abrasives used during ship hull cleaning, degradation of paints and coatings on ship hulls, and cargoes of cargo-carrying ships falling into the marine environment are among the sources of ship-based MP pollution in marine [48]. It is worth noting that, currently, the effect of marine-based sources on MP pollution in marine is less investigated than the effect of land-based sources.

Table 1
Abundance and dominant characteristics of MPs in marine waters.

	Number of samples	Abundance	Dominant polymer type of MP	Dominant shape of MP	Dominant size of MP (mm)	Reference
Black Sea	23	Lowest: - Highest: 16.30 ± 1.20 Mean: 7.30 ± 4.90 (particles/L)	-	Fiber	0.101-0.5	[54]
Andaman Sea	15	Mean: 0.27 ± 0.18 (particles/m^3)	PP	Fragment	> 1	[55]
Black Sea	23	Lowest: 4.42 ± 1.34 Highest: 55.67 ± 9.7 Mean: 18.68 ± 3.01 (particles/m^3)	PET	Fiber	Lowest: 0.23 Highest: 4.98 Mean: 2.07 ± 0.04	[49]
Marmara Sea	9	Mean: 276.18 (particles/km^2) (marine stations)	-	Fragment	1-5	[50]
Marmara Sea	8	Mean: 3497.02 (particle/km^2) (piers)	-	Fragment	0.3-1	[50]
Western Pacific	8	Lowest: 0.02 Highest: 0.10 Mean: 0.06 ± 0.03 (particles/m^3)	Epoxy resin	Granule	-	[56]
South China Sea	6	Lowest: 0.05 Highest: 0.26 Mean: 0.13 ± 0.07 (particles/m^3)	PP	Granule	-	[56]
Eastern Indian Ocean	36	Lowest: 0.01 Highest: 4.53 Mean: 0.34 ± 0.80 (particles/m^2)	PP	Fragment	< 1	[57]
Sea of Japan	4	Mean: 77.50 ± 12.02 (particles/L)	PET	Fiber	2-5	[58]
Seto Island Sea	14	Mean: 53.82 ± 20.26 (particles/L)	PET	Fiber	1-5	[58]
Nordic Sea (affected by the Greenland Sea Gyre)	9	Mean: 2.43 ± 0.84 (particles/L)		Fiber	0.1-0.5	[59]
Nordic Sea (affected by the East Greenland Current)	11	Mean: 1.19 ± 0.28 (items/L)	-	Fiber	0.1-0.5	[59]
Black Sea	-	Lowest: 1.14×10^4 Highest: 1.91×10^5 (particles/km^2) Mean: 4.62×10^4 (particles/km^2)	-	-	-	[60]
Mediterranean Basin	3	Lowest: 2.35 Highest: 6.70 (particles/m^3)	PE	Fragment	< 800 (µm)	[61]
Northwestern Pacific	2	Lowest: 0.018 Highest: 0.035 Mean: 0.030 ± 0.017 (particles/m^3)	PET	Fiber	Mean: 1.73 ± 1.06	[5]
Bering Sea	5	Lowest: 0.035 Highest: 0.26 Mean: 0.091 ± 0.094 (particles/m^3)	PET	Fiber	Mean: 1.57 ± 0.89	[5]
Chukchi Sea	6	Lowest: 0.086 Highest: 0.31 Mean: 0.23 ± 0.07 (particles/m^3)	PET	Fiber	Mean: 1.39 ± 0.74	[5]
Northwestern Pacific	18	Lowest: 640 Highest: 42213 Mean: 187982 (particles/km^2)	PE	Granule	0.5-1	[62]
South China Sea	30	Lowest: 1400 Highest: 8100	PP	Pellet	< 0.5	[63]

(*continued on next page*)

Table 1 (*continued*)

	Number of samples	Abundance	Dominant polymer type of MP	Dominant shape of MP	Dominant size of MP (mm)	Reference
		Mean: 4933 ± 1369 (particles/m^3)				
Yellow Sea	50	Lowest: 0 Highest: 0.81 Mean: 0.13 ± 0.20 (particles/m^3)	PE	Fragment	0.35-44.99 Mean: 3.72 ± 4.70	[64]
Ionian Sea	30	Lowest: - Highest: 1.61 Mean: 0.18 (< 1 mm) Mean: 0.23 (1-5 mm) (particles/m^2)	PE	Fragment	1-5	[65]
Black Sea	3	Mean: 2.66 ± 2.32 particles/m^3 (surface) Mean: 24.47 ± 26.15 particles/m^3 (water column)	-	-	-	[51]
Mediterranean Sea	17	Lowest: 16339 Highest: 520213 (particle/km^2)	-	-	-	[66]
Black Sea		Mean: 1.20 ± 1.10 × 10^3 (particles/m^3) (November) Mean: 0.60 ± 0.55 × 10^3 (particles/m^3) (February)	-	-	-	[67]

3. Abundance of microplastics in marine waters

The abundance of MPs in the marine environment varies depending on the population, environmental awareness of the population, socioeconomic status of the population, industrial activities, number of wastewater treatment plants, MP removal efficiency of wastewater treatment plants, maritime traffic, environmental factors (rainfall, wind, wave movements, etc.). Therefore, MP abundance in the marine environment can significantly show the effect of a single dominant factor or a combination of multiple factors. For instance, in the study conducted by Terzi et al. (2022), they divided the Turkish coast of the Black Sea into four regions: eastern (8 sampling points), western (7 sampling points), middle (4 sampling points) and Marmara (4 sampling points). They investigated the abundance of MP in seawater samples collected from a total of 23 different sampling points. In the study, MP abundance was found to be higher in the Marmara Region, which is the most populous and industrialized region of Turkey. However, although the population and industrial activities in the eastern Black Sea are lower than those in the middle Black Sea, the second region with the highest MP pollution was determined as the eastern Black Sea. This is associated with the fact that the eastern Black Sea region receives much more precipitation during the year [49].

MP abundance in the marine environment is significantly affected by spatial conditions. In a recent study by Sari Erkan et al. (2021), MP analysis was carried out on water samples collected from 9 marine stations and 8 piers in the Marmara Sea. In the study, it was determined that MP pollution was at much higher levels due to intense ship activity and anthropogenic effects at the piers. MP abundance was reported to be lower in marine stations, as marine stations are parking lots for yachts and more elite areas used for marine sports [50]. In addition to the region where the water sample was collected from the marine environment, the depth from which the sample was collected from the marine environment (surface, water column, and close to the seafloor) and the distance to the shore also affect the abundance of MPs [51].

MPs whose density is lower than seawater tend to float, while MPs whose density is higher than seawater tend to sink. However, even though MPs tend to float when they enter the marine environment, they can sink due to the formation of a biofilm layer on their surfaces and the accumulation of different pollutants on their surfaces [52]. Depending on the properties of the MPs, environmental conditions, and microorganisms and pollutants on the MP surface, sinking may occur after several weeks [53]. On the contrary, MPs with higher density than seawater can be present on the surface of seawater where there is wavy and turbulent mixing [22].

Table 1 shows the abundance and dominant characteristics (polymer type, shape, size) of MPs in marine waters. Currently, there is no standard method for the analysis of MPs in water. Different sampling, extraction, and analytical techniques are used by researchers to determine the abundance of MPs in marine waters. The lack of a standard MP analysis method makes direct comparisons of results in the literature very complex. Additionally, since there is no standard procedure for MP analysis, it may lead to an underestimation or overestimation of the MP abundance in the marine environment. For instance, failure to perform good chemical digestion for MP analysis may lead to an overestimation of MP abundance in the water sample. In addition, the abundance of MPs in the marine environment reported by researchers covers MPs in different size ranges. Another consequence of the lack of a standard sampling method is that researchers report MP abundance in different units. MP abundance is reported in different units such as particles/L, particles/m^3, particles/m^2, and particles/km^2 (Table 1). Since some units are not comparable with each other, it is not possible to directly compare the results with each other. Worldwide acceptance of a standard MP analysis method enables direct comparison of research studies with each other. Additionally, a standard MP analysis method helps to better understand the abundance and sources of MPs and environmental factors affecting MP abundance in marine environment.

4. Characteristics of microplastics in marine waters

4.1. Polymer types of microplastics

Identification of polymer types of MPs in seawater may be useful to predict their potential sources. The most used method by researchers for the detection of polymer types of MPs is Fourier Transform Infrared (FTIR) spectroscopy. It has been reported in many studies that MPs such as PE, PP, PA, polystyrene (PS), polyurethane (PU), polyvinyl chloride (PVC), polyethylene terephthalate (PET), polycarbonate (PC), polylactic acid (PLA), acrylonitrile-butadiene-styrene copolymer (ABS), polymethylmethacrylate (PMMA), polyacrylonitrile (PAN), polyvinyl acetate (PVAC), polyvinyl alcohol (PVA) and nitrile butadiene rubber (NBR) are found in marine waters [49,61,65,68]. PE [61,64,65,68] and PP [55,56,63] have been reported as the dominant polymer types of MPs in marine waters around the world. The fact that PE is the most widely produced and used polymer type in the world is one of the reasons why PE MPs are abundant in marine waters [69]. Many items used in daily life, such as the packaging of food, beverages, personal care products and cleaning agents, and shopping bags, are made of PE [3]. Additionally, PE-type MPs are commonly used in microbeads in personal care products [31]. Similarly, PP is widely used in packaging, industrial plastic parts production, and textile products due to its superior properties. Another reason why PE and PP types of MPs are abundant on the surface of sea water, in addition to their widespread use in the world, is that their density is lower than the density of sea water. The densities of low-density PE (LDPE), high-density PE (HDPE), PP, and seawater are 0.89–0.93, 0.94–0.98, 0.85–0.92, and 1.025 g/cm^3, respectively [70]. Therefore, it is not surprising that PE and PP MPs, which have low density and tend to float in seawater, are frequently encountered. However, polymer types of MPs with lower density than seawater (PE, PP, PS, etc.) are found in marine sediments [61,68,71], and polymer types of MPs with higher density than seawater (PET, PA, PVC, PU, etc.) are found on the water surface and in the water column [5,49,58,68]. As mentioned before, the density of MPs depending on the polymer type, environmental conditions in the marine environment and biofilm layer/other pollutants accumulated on the surface of MPs affect the abundance of MPs and the polymer distribution of MPs in the water sample.

Additionally, the polymer types of MPs detected in seawater are related to the water depth at which the water sample was collected and the collection method. For instance, PE (24%), PU (23%), PVAC (12%), PS (10%), PP (10%), PVC (9%), PMMA (7%), rubber (2%), PVA (2%) and alkyd varnish (1%) polymer types were detected in MPs in water samples collected horizontally from sea surface water in Rijpjorden-Northern Svalbard [72]. In the same study, alkyd varnish (40%), PU (40%), PMMA (13%), and PE (7%) polymer types of MPs in water samples collected vertically from the water column (depth: 0–200 m) [72]. However, since water samples collected vertically from the water column include the water surface, the PE-type MPs detected in the water column may have also been captured from the water surface. Therefore, density is a weak factor in the abundance of polymer types of MPs in samples collected vertically from the water column.

As a result, the distribution of polymer types of MPs in marine waters and marine sediments, which are still being investigated by researchers in different parts of the world, is affected by the sources of MPs, the properties of MPs, environmental conditions, and sampling methods. Since more than one factor is effective in the distribution of MPs in the marine environment, the factors affecting the polymer distribution of MPs in marine waters and marine sediments cannot be determined with certainty.

4.2. Shapes of Microplastics

Many studies have reported that the shapes of MPs in marine waters include fragments, fiber, films, granules, pellets, and foams. Shape as well as polymer type can also help predict the potential source of MPs in the marine environment. Fragment-shaped MPs are generally formed by the breakdown of large-sized and thicker plastics than films through various physical, chemical, and biological effects. Film-shaped MPs are formed by breaking thin plastics, such as grocery bags, into smaller pieces. Fibers have mainly two sources in the marine environment. The first of these is the direct discharge of abundant fiber-shaped MPs, which pass into the wastewater when washing synthetic clothes and textiles, into the receiving environment and/or the discharge of the effluent of WWTPs into the receiving environment. Many researchers have revealed that the fiber is the dominant MP shape in the effluent of WWTPs [3]. The second reason for the presence of fiber-shaped MPs in the marine environment is that fishing equipment left/fallen into the marine environment breaks down into smaller pieces due to abrasion, wear, and tear as a result of exposure to various effects over time. A study by Welden and Cowie (2017) showed that PE, PP and nylon ropes exposed to benthic conditions at a depth of 10 m lost approximately 0.45%, 0.39%, and 1.02% mass per month, respectively [73]. It is worth noting that environmental conditions as well as resources can influence the shape of MPs in the marine environment. For instance, ropes and nets dragged on rocky seafloor wear out more than those dragged on sandy seafloor, and this may increase the proportion of fibers in the MP shape distribution in rocky areas where fishing activities are carried out [47].

Different MP shapes such as fragment, fiber, film, and pellet are found together in marine and beach sediments. As a remarkable result, recent studies have reported that fiber and fragment-shaped MPs are dominant in marine or beach sediments [49,61,65,68,71,74], just as in seawater. Most of the studies have revealed that especially fragments and fibers are the dominant MP shape in marine waters (Table 1). While some studies have shown that pellets are the least common among the MP shapes in the marine environment [50,68], some studies have shown that pellet-shaped MPs are the dominant MP shape in the marine environment [63].

Additionally, the shape of MPs in the marine environment can provide an imprecise estimate of their marine residence time. Generally, MPs with sharp corners are new to the marine environment, while MPs with smooth corners can be estimated to have existed in the marine environment for a long time. However, it should be noted that spherical microbeads (primary MPs) used in personal care products also have smooth surfaces. Therefore, shape, color, and FTIR spectrum of the polymer should be evaluated together to estimate the residence time of MPs in the marine environment. Because shifts occur in the FTIR peaks of aged MPs that have been in the environment for a long time. The color of MPs that have been in the marine environment for a long-time fades and their sharp corners are replaced by shaved, smoother corners.

4.3. Sizes and colors of microplastics

MPs are smaller than 5 mm in size. In the marine environment of MPs, they break down into smaller-sized MPs and nanoplastics over time, depending on the properties of the MPs and environmental conditions. As the size of MPs decreases their characterization becomes increasingly difficult. For this reason, currently, the number of studies on nanoplastics in marine waters is less than on MPs in marine waters. Size can affect the degradation rate and fate of MPs [50]. Since the size of the plastics affects their permanence in the marine, the permanence of MPs in the marine is probably less than that of a large piece of plastic [50].

The size of MPs in the marine environment is important as it affects their adsorption capacity for other pollutants in the marine environment. Since the surface area of MPs increases as their sizes decrease, small-sized MPs can adsorb more pollutants [3]. Some studies have shown that the adsorption of pharmaceuticals, [75] bactericides [76], and arsenic [77] to MP increases with decreasing MP size. Therefore, the ingestion of smaller-sized MPs by marine organisms means the transfer of a greater amount of adsorbed pollutants to marine organisms. Considering the fact that plastics in the marine environment break down into smaller and smaller particles, the threat to marine organisms is unfortunately increasing.

Just like shape, color can also be an indicator for estimating the residence time of MPs in the marine environment. While bright and vibrant tones indicate that MPs have recently entered the marine environment, pale and dull tones may indicate that MPs have spent a long time in the marine environment [62]. It has been reported in many studies that MPs are found in transparent, white, black, and colored (blue, green, red, yellow, etc.) colors in marine waters [5,51,57]. However, marine organisms ingest MPs similar in color to their prey, and thus simultaneously take in plastic, various chemicals, and pollutants (microorganisms, pharmaceuticals, PAHs, PCBs, heavy metals, etc.).

5. Negative effects of microplastics on marine organisms

Primary and secondary MPs in the marine environment ingested by marine organisms cause various dysfunctions and problems. Studies have proven the existence of MPs in various marine organisms (Table 2). Ways of MP uptake by marine organisms include the following: **(1)** ingestion of MPs that resemble their prey by marine organisms, **(2)** ingestion of prey that has previously taken MP, **(3)** ingestion of MPs by organisms that feed by filtering the water, and **(4)** uptake of MPs by organisms living in the benthic zone during sediment mixing. All marine organisms, from low to higher trophic levels in the marine environment, are exposed to MPs, which are abundant on the sea surface, in the seawater column, and marine sediments, that is, everywhere in the marine environment. Table 2 presents the occurrence frequency, abundance, and dominant characteristics of MPs in various marine organisms.

Chemical substances such as plasticizers, flame retardants, and dyes contained in MPs [78] also cause harm to marine organisms. MPs adsorb many pollutants in water due to their small size, large surface area, and hydrophobic structure [3] and they can adsorb heavy metals [79], polychlorinated biphenyls (PCBs) [80], polyaromatic hydrocarbons (PAHs) [81] and pharmaceuticals [82] in water. Therefore, MPs, chemicals contained in MPs, and pollutants adsorbed on the surface of MPs can cause functional disorders in marine organisms. In the following sections negative effects of MPs in phytoplankton, zooplankton, corals, fish, sea turtles, seabirds are examined. Table 3 summarizes the negative effects of MPs on various marine organisms.

5.1. Phytoplankton

MPs cause multiple adverse effects on phytoplankton, which are the primary producers of the marine ecosystem. It is known that the negative effects of MPs on phytoplankton include inhibition of cell growth, decrease in photosynthesis ability, decrease in chlorophyll content, formation of hetero-aggregates, inhibition of biomass productivity, restriction of interaction with the environment as a result of adsorption to the surface, and changes in gene expression [90–93]. In the study of Zhang et al. (2017), it was reported that the chlorophyll content of marine microalgae *(Skeletonema costatum)* decreased by 7% at 72 h and 96 h as a result of exposure to 5 mg/L PVC MPs. As a result of exposure of microalgae to 50 mg/L PVC MPs, chlorophyll content decreased by 20% from 24 h to 96 h. In the study, it was determined that increasing PVC MP concentration caused a decrease in algae density and a decrease in photosynthesis ability, especially in the first exposure phase [91]. It is known that the negative effects of MPs on phytoplankton, as well as on other marine organisms, vary depending on the concentration and properties of MPs in the environment. Sjollema et al. (2016) found that as a result of 72 h exposure of marine flagellate *(D. tertiolecta)* to commercially purchased PS particles of 0.05 μm size (250 mg/L), the cell density and algal growth rate of D. tertiolecta decreased by 45% and 45%, respectively. Exposure to 0.5 μm sized PS particles caused a smaller reduction (11%) in the growth of *D. tertiolecta*, whereas 6 μm sized PS MPs caused less than a 10% adverse effect even at low concentration (25 mg/L). The results revealed that decreasing PS particle size further increased the negative effect on the growth rate of microalgae [93]. Another recent study by Fu et al. (2019) revealed that aged PVC MPs inhibited microalgal growth of *Chlorella vulgaris* more strongly than virgin PVC MPs.

Since different conditions prevail in the real marine environment than laboratory conditions, and there are many MPs in different polymer types, sizes and with different properties (aged, charged, uncharged, etc.), the negative effects on marine phytoplankton and other marine organisms can be different. It is worth noting that MPs in real marine environments have different physical and chemical structures than commercially available MPs used in laboratory studies. In the marine environment, changes in the physical and chemical structures of MPs may occur as a result of exposure to various physical, chemical, and biological effects. Compared to virgin MPs, aged MPs contain more cracks and pores in their structures, their surface areas are larger and their functional groups are different [94,95]. Since virgin and aged MPs have different properties, they have different negative effects on phytoplankton and other marine organisms.

5.2. Zooplankton

Zooplankton are small aquatic animals that are the primary consumers of the marine environment, and they feed by consuming phytoplankton [96]. Cole et al. (2013) reported that 13 zooplankton taxa, which are common in the northeast Atlantic and include holoplankton, meroplankton, and microzooplankton, can ingest PS MPs < 31 μm in size in the absence of natural food. The copepod *(P. annandalei)*, which feeds on microalgae, showed the highest lipid production (17.3%), while this rate was found to be lower in the copepod that ingests MP [98]. Additionally, while the maximum survival rate of copepods fed with *Isochrysis galbana* was 80%, copepods that ingested MP died on the 12th day and the survival rate was determined as 0% [98]. In the study of Xuekai et al., (2021), as a result of exposure of brine shrimp *(Artemia parthenogenetica)* to 2 mg/L PS MPs at 30 °C, it was reported that the growth and survival rate decreased significantly, the intestinal microvilli and epithelia cells of Artemia decreased, the intestinal tissue was negatively affected, and immunological stress and oxidative stress were induced. Similarly, in a recent study by Li et al. (2021), brine shrimp *(Artemia parthenogenetica)* were exposed to seawater, seawater containing 100 mg/L PE MPs and seawater containing 100 mg/L PS MPs. The results revealed that MPs decreased the growth rate and body length of *Artemia* and caused a serious disorder in its gut microbiota [100].

MPs ingested by zooplankton are transferred to higher trophic levels through the predator-prey relationship. In addition to the negative consequences observed in zooplankton as a result of exposure to MPs, the fact that MPs reduce the survival rate of zooplankton and reduce lipid production brings about the problem of food and energy deficiency for marine organisms at higher trophic levels that consume zooplankton. Additionally, MPs ingested by zooplankton are excreted within their faecal pellets. MPs can significantly change the sinking rate, density and structural integrity of faecal pellets egested by zooplankton [101]. These pellets, which are vectors for MPs, are a food source for the marine organism and when consumed, they are transferred to different organisms and cause various problems.

Table 2
Occurrence frequency, abundance, and dominant characteristics of MPs in marine organisms.

Marine organisms		Location	Occurrence frequency (%)	MPs abundance (particles/ individual)	Dominant characteristics of MPs	Reference
Crabs (n = 120)	*Callinectes sapidus*	Western Mediterranean Sea	65.80	1.4 ± 1.6	0.5-1 mm, fiber, blue, PE	[83]
Seabirds (n = 45)	*Calonectris borealis*	Tafira Wildlife Rehabilitation Center, Gran Canaria	88.89	7.22 ± 5.66	Line, green and blue	[84]
Seabirds (n = 5)	*Oceanodroma castro*	Tafira Wildlife Rehabilitation Center, Gran Canaria	100	5.60 ± 2.88	Fragment, yellow and white	[84]
Seabirds (n = 20)	*Larus michahellis*	Tafira Wildlife Rehabilitation Center, Gran Canaria	30	0.55 ± 1.05	Line and film, green	[84]
Seabirds (n = 2)	*Chroicocephalus ridibundus*	Tafira Wildlife Rehabilitation Center, Gran Canaria	50	0.50 ± 0.71	Film, red	[84]
Seabird (n = 1)	*Bubulcus ibis*	Tafira Wildlife Rehabilitation Center, Gran Canaria	100	6	Line, transparent	[84]
Cultured clams (n = 34)	*Ruditapes philippinarum*	Jiaozhou Bay	82.86	2.28 ± 1.33	< 500 µm, fiber, black, cellophane	[85]
Wild clams (n = 20)	*Ruditapes philippinarum*	Jiaozhou Bay	65	1.38 ± 0.87	< 500 µm, fiber, blue, cellulose and cellophane	[85]
Cultured oysters (n = 33)	*Crassostrea gigas*	Jiaozhou Bay	57.58	1.53 ± 0.84	< 500 µm, fiber, black, PET	[85]
Wild oysters (n = 33)	*Crassostrea gigas*	Jiaozhou Bay	75.76	2.96 ± 2.09	< 1500 µm, fiber, black, cellophane	[85]
Shrimps (n = 60)	*Metapenaeus monoceros*	Northeast Arabian Sea	100	7.23 ± 2.63	100-250 µm, fiber, black and red	[86]
Shrimps (n = 50)	*Parapeneopsis stylifera*	Northeast Arabian Sea	100	5.36 ± 2.81	100-250 µm, fiber, black and red	[86]
Shrimps (n = 70)	*Penaeus indicus*	Northeast Arabian Sea	100	7.40 ± 2.60	100-250 µm, fiber, black and red	[86]
Fish (n = 14)	*Chelon saliens*	Southern Caspian Sea	-	4.2	500-1000 µm, fiber, black-grey	[87]
Fish (n = 10)	*Cyprinus carpio*	Southern Caspian Sea	-	3.4	500-1000 µm, fiber, black-grey	[87]
Fish (n = 11)	*Rutilus caspicus*	Southern Caspian Sea	-	1.6	500-1000 µm, fiber, black-grey	[87]
Zooplankton	*Medusae*	Bohai Sea	-	0.056 and 0.117	-	[88]
Fish (n = 30)	*Engraulis encrasicolus*	Eastern Mediterranean Basin	83.40	2.5 ± 0.3	< 400 µm, fragment, PS	[61]
Spiny oysters (n = 30)	*Spondylus spinosus*	Eastern Mediterranean Basin	86.30	7.2 ± 1.4	< 400 µm, fiber, PS	[61]
Fish (n = 80)	*Sardina pilchardus, Pagellus erythrinus, and Mullus barbatus*	Northern Ionian Sea	41.25	1.66 (< 1 mm) 0.06 (1-5 mm)	< 1 mm, fragment, PE	[65]
Mussels (n = 80)	*Mytilus galloprovincialis*	Northern Ionian Sea	46.25	1.83 (< 1 mm) 0 (1-5 mm)	< 1 mm, fragment, PE	[65]
Fish (n = 1337)	28 species	Mediterranean Sea	58	2.36	Fibers and nylon	[66]
Sea turtles (n = 24)	*Caretta caretta*	North Atlantic subtropical gyre	58 (1-5 mm large MPs)	-	Fragment	[89]

5.3. Corals

MPs are also present in coral reefs, which are the habitats of marine creatures and have rich biodiversity [102,103]. A study in Lombok, Indonesia, showed the presence of MPs in sediments collected from all sampling points in 10 coral reef habitats (35–77 particles/kg), with an average MP abundance of 48.3 ± 13.98 particles/kg [104]. In 10 different stations, the dominant MP shape was foam (41.20%), the dominant MP polymer type was PS (41.20%) and the dominant MP size was > 1000 µm (41.61%). In the study, considering the polymer types and shapes of MPs, it is estimated that MPs in coral reef sediments originate from fishing gear, styrofoam, and food and beverage packages [104]. In the marine environment, MP cause many negative effects on corals. Corals exposed to MPs respond differently, including ingestion, mucus production, outgrowth, and attachment of MPs to tantacles or mesenterial filaments [105]. A study by Chen et al. (2022) showed that PE MPs adsorbed on the surface of corals *(Goniopora columna)* and could enter the tissues of corals. The research showed that the number of MPs both on the surface and inside of the coral increased with increasing PE MP concentration from 5 mg/L to 300 mg/L, with exposure for up to 7 days. For 300 mg/L PE MP concentration, the number of MPs on the surface and inside of the coral was determined to be 992.5 ± 12.0 and 26.5 ± 0.7, respectively. As a result of short-term exposure of *Goniopora columna* to MPs, MPs reduced polyp length, reduced adaptation, inhibited the function of enzymes responsible for detoxification, caused zooxanthellae density changes, interfered with the anthozoan-algae symbiotic relationship, produced excess lipid peroxide, and led to tissue damage [106]. Additionally, other adverse effects of MPs on corals include reduced skeletal growth rate [107], tissue necrosis, and bleaching [105].

Table 3
Negative effects of plastics and microplastics on marine organisms.

	Negative Effects of MPs on Marine Organism	Reference
Phytoplankton	• Aggregate formation • Decrease in biomass productivity • Decrease in cell density • Decrease in chlorophyll content • Decrease in CO_2, light, and nutrient uptake • Decrease in growth • Decrease in photosynthesis efficiency • Limitation of interaction with the environment • Promotion of reactive oxygen species (ROS) production	[90],[92],[93],[98],[91],[119]
Zooplankton	• Change in gut microbiota • Changes in reproduction • Decrease in algal ingestion rate • Decrease in body length • Decrease in lipid concentration • Decrease in nauplii production rate • Decrease in survival rate • Oxidative and immunological stress • Reduction and inhibition of growth	[97],[98],[99]
Corals	• Bleaching • Decrease in adaptation • Decrease in polyp length • Excessive production of lipid peroxide • Inhibition of the function of enzymes responsible for detoxification • Intervention in symbiotic relationship • Reduction in skeletal growth rate • Tissue damage and tissue necrosis • Zooxanthellae density changes	[105],[106],[107]
Mussels	• Decrease in filtration rate • Inactivation of the antioxidative system • Induction of oxidative damage/oxidative stress	[120],[121]
Oysters	• Change in diet • Change in lipid-glucose metabolism processes • Decrease in larval development • Induction of oxidative stress • Negativities in growth in the larval stage • Reproductive disorder	[15],[122]
Sea turtles	• Change in reproductive success • Changes in nest temperature on beaches • Gender imbalance	[113],[114]
Fish	• Changes in swimming behavior • Decrease in body length • Decrease in growth rate • Decrease in head/body ratio • Decrease in mass • Decrease in survival rate, • DNA breaks • Histopathological lesions in the liver and intestine • Immune system dysfunction • Inflammation • Mortality • Oxidative damage to the liver	[13],[14],[108],[123],[124],[125]
Sea birds	• Blockage of the gastrointestinal tract • Change in gut microbiomes • Decrease in body weight and/or fat load • Decrease in feeding and energy • Decrease in stomach capacity • False satiation and decrease in energy • Induction of multi-organ damage • Internal injuries • Mortality • Ovulation delay • Puncture of the gastrointestinal tract • Reproductive failure	[115],[116],[117],[126]

5.4. Fish

Studies conducted in recent years have reported the abundance and characteristics of MPs in many marine and freshwater fish. For instance, a study conducted in Turkish territorial waters in the Mediterranean Sea showed that MPs were present in the stomach of 34%, in the intestine of 41%, and in the stomach and/or intestine of 58% of 1337 fish representing 28 species and 14 families [66]. Considering only fish that had ingested MP, the abundance of MP in the stomach, intestine, and stomach and/or intestines of fish was found to be 1.80, 1.81, and 2.36 per individual, respectively. The majority of ingested particles were reported to be fiber-shaped (70%) and blue [66].

MP uptake by fish can occur by direct ingestion of MPs from seawater or indirectly by predation of zooplankton that have ingested MPs at a lower trophic level. MPs can make it difficult for fish to distinguish and effectively capture zooplankton in the marine environment [108].

Plastics and MPs can accumulate in the gastrointestinal system of fish and prevent the digestion of food, cause false satiation, reduce the feeding rate, and cause mortality [109] Depending on their size, swallowed plastics can cause blockage in the digestive system and cause direct mortality [110]. As a result of exposure of juvenile fish to MPs, growth-related negativities such as a decrease in body length, body depth, and fish mass occur, and their survival probability decreases [13]. The decrease in fish population is an important ecological problem. However, it is worth underlining that this problem can cause both ecological and economic problems for fish with economic value. As a result, studies emphasize that the negative effects of MPs on fish include decreased survival and growth, changes in body condition, oxidative stress/oxidative damage, inflammation, and depression of the immune system (Table 3).

5.5. Sea turtles

The life of sea turtles is threatened by MP pollution in the sea and on beaches. Pham et al. (2017) reported that all 380 debris identified in sea turtles *(Caretta caretta)* were plastic, and these plastic pieces mainly consisted of user plastics (15.75 ± 6.59 items/ind), fisheries-related plastics (3.00 ± 1.42 items/ind), and industrial plastics (0.25 ± 0.25 items/ind), respectively. In the study, it was determined that the majority of user plastics, fishing plastics, and industrial plastics consist of fragments (67.6%), synthetic rope (63%), and pellets (100%), respectively. Analysis results revealed that PE, PP, PE + PP, rayon, PVC, PVAC, and nylon polymers were found in sea turtles, and the most abundant polymers were PE (60%) and PP (20%), respectively. It was found that the average debris in the intestines of sea turtles was higher than that in the esophagus and stomach [89]. A study published in 2004 reported the ingestion of plastic pieces by two sea turtles *(Lepidochelys olivacea* and *Chelonia mydas)* [111]. As a result of the autopsy performed on the dead sea turtle *(Lepidochelys olivacea)*, it was determined that there were a total of 10 pieces of plastic in the stomach of the turtle. On the other hand, it was observed that the stranded alive sea turtle *(Chelonia mydas)* defecated a total of 20 pieces of plastic, including 11 pieces of hard plastic and 9 pieces of plastic bags, on the 13th and 14th days [111].

Sea turtles spend almost all their lives in the sea and come to the beach for a short period to lay eggs. However, unfortunately, MPs on beaches have a negative effect on sea turtle eggs. A study conducted in China showed that the average MP abundance in beach surface sediments in the largest nesting area (Qilianyu, northeastern Xisha Islands) of green sea turtles *(Chelonia mydas)* was 338.44 ± 315.69 thousand pieces/m^3 (1353.78 ± 853.68 pieces/m^2) [112]. In addition to MP pollution in beach surface sediments, MPs were also detected in the nest bottoms of sea turtles at a depth of approximately 60 cm. In the study, it was reported that the majority of MPs identified in the sampling areas were PE and PS polymer types, foam and fragment shapes, 0.05–1 mm in size, and transparent-white in color [112]. (The dominance of small-sized MPs at the bottom of sea turtle nests may lead to decreased hatching success or adversely affect embryonic development. The reason for this is that especially small-sized MPs can adsorb more pollutants, and the adsorbed harmful substances have the potential to reach sea turtle eggs through migration [112]. Moreover, the specific heat capacity of MPs is higher than sand, and therefore, MPs, which are abundant on beaches, cause the temperature of the sand to increase further, causing temperature changes around the nests of sea turtles [113]. This can affect the sex ratio of sea turtles, leading to gender imbalance, as well as negatively affecting reproductive success [113,114].

5.6. Seabirds

Seabirds feed on fish and other marine organisms. Seabirds can ingest MPs together with seawater while ingesting their prey. Another way for seabirds to ingest MPs is by ingesting prey at lower trophic levels that have previously ingested MPs (i.e. food chain). Seabirds can directly or indirectly ingest MPs from a wide variety of areas in the marine environment during their migration. A newly published study by Navarro et al. (2023) showed that MPs are found in %88,89, %100, %30, %50 and %100 of Cory’s shearwaters’un *(Calonectris borealis)* (n = 45), madeiran storm-petrels’ın *(Oceanodroma castro)* (n = 5), yellow-legged gull *(Larus michahellis)* (n = 20), black-headed gull’ın (*Chroicocephalus ridibundus)* (n = 2) ve cattle egret *(Bubulcus ibis)* (n = 1), respectively. In the study, the average MP abundance in Cory's shearwaters, madeiran storm-petrels, yellow-legged gulls, black-headed gulls, and cattle egrets was 7.22 ± 5.66, 5.60 ± 2.88, 0.55 ± 1.05, 0.50 ± 0.71 and 6 per individual, respectively. The majority of MPs in Cory's shearwaters were lines (75.08%), and in Madeiran storm petrels were fragments (78.57%). In yellow-legged gulls, lines (36.34%) and films (36.34%) were equally distributed and constituted the majority. While an MP film was detected in only one of the black-headed gulls, 6 lines were detected in the cattle egret. In the study, 34% of the items were analyzed by FTIR, and the dominant MP type was determined to be PE (70.87%), followed by PP (15.75%) [84].

Plastics and MPs ingested by seabirds cause blockage, disruption, and puncture of their digestive system [115]. Additionally, MPs also lead to decreased stomach capacity, nutrition and energy, internal injuries, and changes in body condition [115,116]. Some studies conducted in recent years have also reported that exposure of seabirds to plastic negatively affects their reproduction [116] and causes multi-organ damage [117].

MPs ingested by seabirds can be regurgitated from the proventriculus or mechanically ground in the muscular gizzard. Mechanically ground MPs are broken into pieces small enough to pass through the intestines of seabirds and be excreted through the cloaca [118]. For instance, MPs were detected in 47% (14/30) of Northern Fulmar (Fulmarus glacialis) faecal precursor samples. It was observed that the majority of the detected MPs were fiber-shaped, blue, and 100–1000 µm in size [118]. Therefore, seabirds also cause MPs from the marine environment to be transported to the terrestrial environment through defecation, and this may negatively affect the health of living things in the terrestrial environment.

6. Conclusion

Studies clearly show that MPs are a serious threat to marine biota and marine ecosystems. Plastic production and consumption are increasing every year, which unfortunately means that more MP is accumulating in marine environments globally every year. Reducing dependence on plastic, reducing single-use plastic consumption, improving plastic waste management, adopting strategies based on recycling plastic waste, increasing funds for cleaning the marine environment from plastics, and supporting scientific research on MP removal methods from marine waters are among the methods of reducing plastic and MP pollution in the marine environment. However, for these methods to be effective in reducing MP pollution in the marine environment, the government, the public, and researchers must take joint action. Since researchers use different MP sampling, extraction, and analysis methods in scientific research, this makes it very difficult to compare MP abundance in studies. A globally accepted standard MP analysis will be the solution to this problem. Since MPs cause many negative effects on marine organisms, the exposure of marine organisms to MPs must be constantly monitored meticulously to prevent the balance in the marine ecosystem from being disrupted and to prevent interactions between marine organisms from being negatively affected.

CRediT authorship contribution statement

Conceptualization, Investigation, Writing - Original Draft, Writing-Review & Editing and Visualization. Seren Acarer Arat contributed 100% to this review article.

Data Availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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